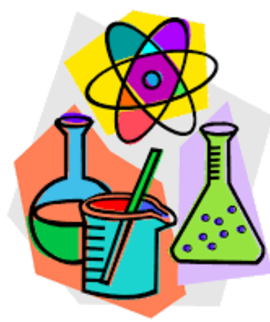
UNIVERSITY INTERSCHOLASTIC LEAGUE

Science

Invitational A • 2022



GENERAL DIRECTIONS:

- **DO NOT OPEN EXAM UNTIL TOLD TO DO SO.**
- Contestants may take up to two hours to complete the contest. If you are in the process of actually writing an answer when the signal to stop is given, you may finish writing that answer.
- Papers may not be turned in until 30 minutes have elapsed. If you finish the test in less than 30 minutes, remain at your seat and retain your paper until told to do otherwise. You may use this time to check your answers.
- All answers must be written on the answer sheet provided. Indicate your answers in the appropriate blanks provided on the answer sheet. Write clearly and legibly!
- You may place as many notations as you desire anywhere on the test paper but not on the answer sheet, which is reserved for answers only.
- You may use additional scratch paper provided by the contest director.
- All questions have ONE and only ONE correct (BEST) answer. There is a penalty for all incorrect answers.
- If a question is omitted, no points are given or subtracted.
- The back two pages of this test include a copy of the periodic table of the elements, as well as listings of other scientific relationships. You may use this information during the contest and may detach the back page from the test if you wish.
- A simple scientific calculator is sufficient for the high school Science contest. **The UIL provides a list of approved calculators that meet the criteria for use in the Science contest. No other calculators are permitted during the contest.** The Science Contest Approved Calculator List is available in the current Science Contest Handbook and on the UIL website. Contest directors will perform a brief visual inspection to confirm that all contestants are using only approved calculators. Each contestant may use up to two approved calculators during the contest.

B01. The endomembrane system of the eukaryotic cell includes all of the following except
A) rough endoplasmic reticulum.
B) Golgi apparatus.
C) nuclear envelope.
D) centrioles.
E) smooth endoplasmic reticulum.

B02. In Mendelian classical genetics, what percent of the offspring from the following genetic cross would be homozygous dominant?

Pp x pp

A) 0%
B) 25%
C) 50%
D) 75%
E) 100%

B03. All of the following are characteristic of the human genome in a somatic cell except
A) linear chromosomes.
B) associated histone proteins.
C) haploid arrangement.
D) multiple chromosomes.
E) housed within a membranous structure.

B04. Blood is the connective tissue most associated with the
A) musculoskeletal system.
B) cardiovascular system.
C) nervous system.
D) integumentary system.
E) digestive system.

B05. Which major macromolecular group contains molecules that store inherited information?
A) Proteins
B) Nucleic acids
C) Carbohydrates
D) Amino acids
E) Lipids

B06. The tissue type that provides covering and lines body cavities is classified as
A) nervous.
B) connective.
C) epithelial.
D) organs.
E) muscle.

B07. Carbon dioxide is an important substrate in photosynthesis. What is the carbon dioxide used for in photosynthesis?
A) Carbon dioxide is released by the photosynthetic organism.
B) Carbon dioxide is used as an electron donor.
C) Carbon dioxide is released during the reactions of aerobic respiration.
D) Carbon dioxide is used as an electron acceptor.
E) Carbon dioxide is used to synthesize sugars, such as glucose.

B08. The major molecular machinery that generates proteins in cells is called the
A) amino acid.
B) rough endoplasmic reticulum.
C) polymerase.
D) ribosome.
E) Golgi apparatus.

B09. In terms of evolution, what do the following terms have in common?

Mutation, recombination, migration, genetic drift

A) These are all events that occur in populations at Hardy-Weinberg equilibrium.
B) These terms are all processes that can happen to proteins.
C) These events may affect genetic diversity in a population.
D) These processes all lead to extinction events.
E) Speciation is always an outcome of these events.

B10. In a population at Hardy-Weinberg equilibrium, which value in the mathematical formula(s) represents the frequency of heterozygous individuals in the population?
A) p^2
B) p
C) q^2
D) q
E) 2pq

B11. What major event is occurring in the S-phase of the cell cycle?
A) Condensation of the chromosomes
B) DNA replication
C) Cellular growth
D) Formation of the spindle apparatus
E) Division of the cytoplasm

B12. Consider the F2 generation of Mendel's pea plants. If 16 progeny were produced in this generation from a dihybrid cross, statistically, how many of the 16 would be homozygous recessive for both traits?
A) 1
B) 3
C) 6
D) 8
E) 16

B13. Hybrids, such as mules (donkey x horse), are sterile because they do not produce functional gametes. This is evidence of
A) prezygotic barriers.
B) horses and donkeys are members of the same species.
C) postzygotic barriers.
D) mutation
E) gene flow.

B14. A DNA binding protein that binds to a sequence located upstream of the promoter and assists RNA polymerase with finding the promoter would most likely be involved in
A) repressing transcription.
B) activating transcription.
C) repressing translation.
D) activating translation.
E) negative control.

B15. Of the four supergroups found in eukaryotic classification, to which do humans belong?
A) Unikonta
B) SAR
C) Excavata
D) Archaeplastida

B16. The Centers for Disease Control and Prevention have posted several *Salmonella* outbreaks from August 2021 through the end of October 2021. Which of the following modes of transmission is most important for *Salmonella* infections?
A) parenteral (needle stick, bee sting, etc)
B) fomites
C) vectorborne
D) ingestion
E) inhalation

B17. Examine the answer choices listed below. Which answer choice is a disease not caused by a eukaryotic pathogen?
A) Hookworm disease
B) Malaria
C) Scabies
D) Amoebic dysentery
E) Chickenpox

B18. Energy-releasing reactions in the cell are
A) anabolic.
B) catabolic.
C) endergonic.
D) exergonic.
E) More than one answer above is correct.

B19. Fleas are often found feeding on the blood of animals. Which kind of organism relationship does this example represent?
A) parasitism
B) mutualism
C) commensalism
D) beneficial
E) reciprocal

B20. There are three domains of life, one of which includes the Archaea. Which of the following *best* represents this domain?
A) Eukaryotic cell types.
B) Single-celled organisms capable of causing disease.
C) Single-celled organisms that are found as part of the human microbiome, particularly in the GI tract.
D) Typically extremophiles (organisms loving extreme environments), including high salt, low or high temperatures, or high pressure.
E) Multicellular organisms capable of producing different cell types and tissues with specific function.

C01. What is the molar mass of sodium hypobromite (NaBrO)?

A) 49.80 g/mol
B) 74.44 g/mol
C) 118.89 g/mol
D) 125.89 g/mol
E) 134.89 g /mol

C02. If 100.0 g of Al reacts with excess oxygen to form Al_2O_3, how many grams of Al_2O_3 will be formed?

A) 179 g
B) 189 g
C) 199 g
D) 209 g
E) 219 g

C03. What is the energy of a photon that has a wavelength of 510 nm?

A) 3.90×10^{-19} J
B) 3.90×10^{-28} J
C) 3.90×10^{-21} J
D) 1.30×10^{-27} J
E) 1.30×10^{-25} J

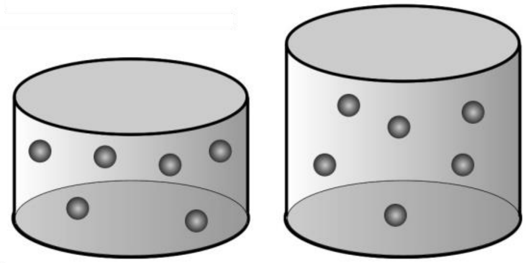
C04. What is the shape of a carbon tetrachloride (CCl_4) molecule?

A) pentagonal
B) tetrahedral
C) square planar
D) trigonal planar
E) linear

C05. Based on periodic trends, which of these ionic compounds would be expected to have the highest melting point?

A) NaCl
B) Na_2O
C) $MgCl_2$
D) MgO
E) K_2S

C06. Containers A and B are at the same temperature. If the pressure in container A is 1.50 atm, what is the pressure in container B?



A) 1.0 atm
B) 1.5 atm
C) 3.0 atm
D) 2/3 atm
E) 6.0 atm

C07. You drop a piece of sodium metal into water and the following exothermic reaction occurs:

$2\ Na(s) + 2\ H_2O(\ell) \rightarrow 2\ NaOH(aq) + H_2(g)$

What are the signs on heat (*q*) and work (*w*) for the system in this reaction, assuming constant pressure and temperature?

A) *q* is positive and *w* is negative
B) *q* is negative and *w* is positive
C) *q* and *w* are both negative
D) *q* and *w* are both positive
E) There is not enough information given

C08. How much energy would it take to melt a pound of ice (454 grams) to water at 0 degrees C?

A) 949 J
B) 1030 J
C) 1890 J
D) 152 kJ
E) 1030 kJ

C09. The reaction $A + 2B \rightleftharpoons C + 2D$ has an equilibrium constant of 1.5×10^{-5}. If the equilibrium concentrations of A, B, and C are 0.0015 M, what is the equilibrium concentration of D?

A) 1.5×10^{-5} M
B) 0.0015 M
C) 1.5×10^{-4} M
D) 2.3×10^{-8} M
E) 5.8×10^{-6} M

C10. Which of these compounds when dissolved in water would result in a solution with a neutral pH?

A) KCl
B) KCN
C) KOH
D) HCl
E) HCN

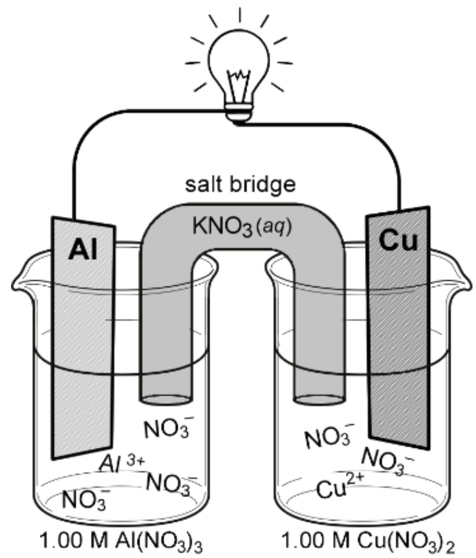
C11. A scientist slowly adds dissolved NaCl to a solution of $AgNO_3$. If a precipitate begins to form when the NaCl concentration is 5.5×10^{-4} M, what is the concentration of Ag^+ in the solution? $K_{sp} = 1.8 \times 10^{-10}$.

A) 3.3×10^{-7} M
B) 9.39×10^{-14} M
C) 3.1×10^{6} M
D) 6.0×10^{-6} M
E) 6.0×10^{-8} M

C12. A tank of compressed helium gas holds 12.8 moles of helium gas. What would the volume of this gas be if it were allowed to expand in a flexible-walled container to 1.00 atm pressure at 25.0°C?

A) 26.3 L
B) 111 L
C) 212 L
D) 313 L
E) 414 L

C13. What is the voltage of this galvanic cell that is made up of a copper cathode in 1.0 M $Cu(NO_3)_2$ solution and an aluminum anode in 1.0 M $Al(NO_3)_3$ solution?



A) +1.32 V
B) –1.32 V
C) +2.00 V
D) –2.00 V
E) +0.34 V

C14. The overall reaction $A + B_2 + C \rightarrow AB + CB$ has the following reaction mechanism:

$A + B_2 \rightarrow AB_2$
$AB_2 \rightarrow AB + B$
$B + C \rightarrow CB$

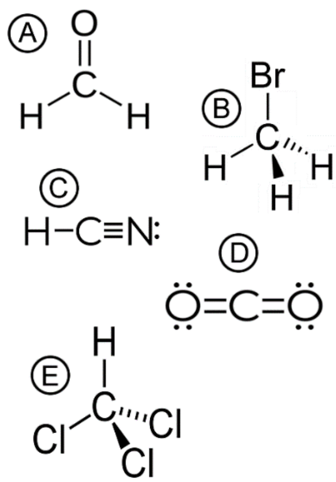
Which compound in the mechanism is a catalyst?

A) A
B) B
C) AB_2
D) B_2
E) There is no catalyst.

C15. Which of these neutral atoms has the smallest radius?

A) Al
B) Br
C) Cl
D) Db
E) Er

C16. In which one of these compounds does the carbon atom have an oxidation state of 0?



C17. How many atoms are in 100. grams of PCl_5?

A) 2.89×10^{23}
B) 1.73×10^{24}
C) 5.78×10^{23}
D) 1.45×10^{24}
E) 6.02×10^{23}

C18. Our current periodic table ends with element 118, which has a full 7*p* subshell in the ground state. What would the last term of the electron configuration be for a ground state atom of element 119?

A) $7p^7$
B) $8s^0$
C) $8s^1$
D) $8s^2$
E) $8p^1$

C19. Compound A has a vapor pressure of 18 torr and compound B has a vapor pressure of 36 torr. Which of the samples below would have the highest vapor pressure?

A) Pure Compound A
B) A mixture with a mole ratio of 0.75 A and 0.25 B
C) A 50/50 mole ratio mixture of A and B
D) A mixture with a mole ratio of 0.25 A and 0.75 B
E) Pure Compound B

C20. A little lost electron walks up to you on the street and says he has fallen out of his atom. He doesn't know what element it was, but he knows he was the outermost electron in the atom. He can remember two of his four quantum numbers: $n = 3$ $\ell = 0$. Five local +1 ions are searching for a lost electron. Which one of these might be this electron's home atom?

A) Carbon
B) Sulfur
C) Potassium
D) Magnesium
E) He could not have come from any of these atoms.

P01. According to Kaku, deep underlying physical principles about the universe are indicated by theories that are ______________.

A) linear
B) non-linear
C) symmetric
D) self-consistent
E) testable

P02. According to Kaku, when William Hershel took a prism, created a rainbow, and placed a thermometer below the visible colors, in a location where there was no color at all, he discovered _____________.

A) magnetic fields
B) electric fields
C) radio waves
D) infrared light
E) ultraviolet light

P03. According to Kaku, acceleration in one reference frame is identical to __________________ in another reference frame.

A) the effect of gravity
B) the slowing of time
C) a constant velocity
D) the electromagnetic force
E) a quantum force

P04. Cooler regions on the Sun, called sunspots, are associated with ________________.

A) strong magnetic fields
B) strong electric fields
C) strong convection currents
D) low neutrino flux
E) a weak solar wind

P05. What is the result of this calculation to the correct number of significant figures?

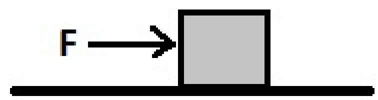
$$X = (6.18)(19.13 - 16.2)$$

A) 20
B) 18
C) 18.1
D) 18.11
E) 18.107

P06. A pinecone drops from a tree onto the forest floor. You estimate that the pinecone was falling for 3.5 seconds. Assuming the pinecone started from rest, and ignoring air resistance, from about what height did the pinecone drop?

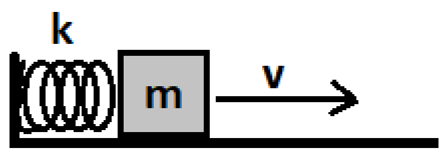
A) 17 m
B) 34 m
C) 60 m
D) 84 m
E) 120 m

P07. A box of rocks having a mass of 210.0kg is pushed along a frictionless floor by a horizontal force of 65.0N (as shown). If the box started from rest, what is its velocity after 2.50 seconds?



A) 8.08 m/s
B) 3.23 m/s
C) 1.93 m/s
D) 1.29 m/s
E) 0.774 m/s

P08. A compressed spring launches a wood block horizontally across a frictionless floor (as shown). The spring has a spring constant of 390 N/m and is initially compressed by 25.0cm. The wood block has a mass of 130g. What is the velocity of the block after it is launched?



A) 9.69 m/s
B) 13.7 m/s
C) 19.4 m/s
D) 27.4 m/s
E) 43.3 m/s

P09. A carousel at a local carnival starts from rest and reaches an angular speed of 2.30 rad/s in exactly one revolution. What is the angular acceleration of the carousel?

A) 0.18 rad/s^2
B) 0.37 rad/s^2
C) 0.42 rad/s^2
D) 0.65 rad/s^2
E) 0.84 rad/s^2

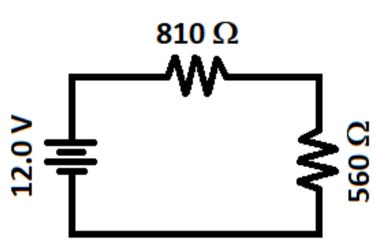
P10. An 85.0kg crate is placed on top of a vertical spring. Once it is released, the crate bounces on the spring with an oscillation period of 2.50s. What is the spring constant, k, of the spring?

A) 130 N/m
B) 210 N/m
C) 250 N/m
D) 400 N/m
E) 540 N/m

P11. A glass window is located in a room maintained at 22.0°C, while the temperature outside is a chilly −14.0°C. The window is 40.0cm wide, 1.30m tall, and 2.50cm thick. How much heat energy is lost through the window each second? The thermal conductivity of glass is 0.80 W/mK.

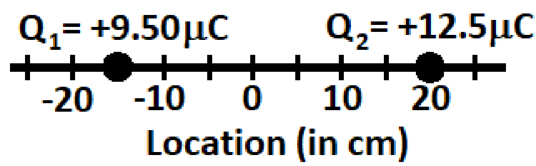
A) 230 J/s
B) 370 J/s
C) 460 J/s
D) 600 J/s
E) 750 J/s

P12. In the circuit below, what is the voltage across the 560 Ω resistor?



A) 2.14 V
B) 4.91 V
C) 7.09 V
D) 8.30 V
E) 12.0 V

P13. Two charges are located on the x-axis as shown. A +9.50μC charge (Q_1) is located at $x = -15.0$cm and a +12.5μC charge (Q_2) is located at $x = +20.0$cm. What is the magnitude of the force on Q_2 due to the presence of Q_1?



A) 3.05 N
B) 5.34 N
C) 8.71 N
D) 17.7 N
E) 26.7 N

P14. A beam of protons is directed into a region of high magnetic field such that the velocity of the beam is perpendicular to the direction of the field. In the field region, the proton beam traces out a circle with a radius of 56.0cm. If the magnetic field strength is 0.60T, then what is the velocity of the protons in the beam?

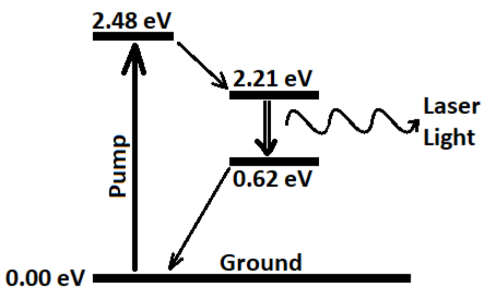
A) 1.93×10^7 m/s
B) 3.22×10^7 m/s
C) 5.37×10^7 m/s
D) 6.44×10^7 m/s
E) 8.95×10^7 m/s

P15. A magnetic field of 0.055 Tesla passes through a square conductive loop. The field is oriented perpendicular to the face of the loop. The square loop is 40.0cm on a side, and has an internal resistance of 1.50Ω. If the magnetic field suddenly vanishes to zero in a time of 0.075 seconds, what current is induced in the conductive loop?

A) 78.2 mA
B) 117 mA
C) 196 mA
D) 293 mA
E) 489 mA

P16. You use a converging lens (a magnifier) to read some very small print. The focal length of the lens is +5.00cm, and you hold the lens 3.50cm above the print. By how much is the print magnified by the lens?
A) 1.70
B) 2.06
C) 2.33
D) 3.33
E) 11.7

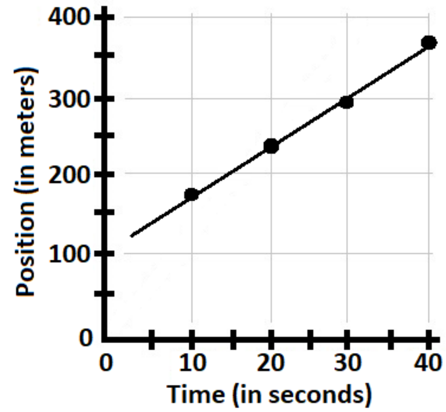
P17. A four-level laser system is set up using atoms that have the energy level diagram shown. What is the wavelength of the laser light produced by this system?



A) 335 nm
B) 500 nm
C) 561 nm
D) 667 nm
E) 780 nm

P18. You have been given an atom of Flerovium-292, $^{292}_{114}Fl$. Very rapidly, the atom emits seven alpha particles, four negative beta particles, and two gamma rays. What atom are you left with after these nuclear decays?
A) Curium-258
B) Fermium-260
C) Rutherfordium-264
D) Nobelium-262
E) Seaborgium-272

P19. The position-time graph for a marathon runner is shown below. Based on this data, what is the approximate speed of the runner?



A) 6.7 m/s
B) 8.8 m/s
C) 10.0 m/s
D) 12.5 m/s
E) 17.5 m/s

P20. You connect a zucchini into an electrical circuit. You then measure the current that flows through the zucchini as you apply different voltages to it. The following table shows the data you collected. Based on this data, what is the approximate resistance of the zucchini?

Voltage	Current
5.0 V	1.33 μA
9.0 V	8.79 μA
12.0 V	14.4 μA
15.0 V	20.0 μA

A) 3.76 MΩ
B) 1.02 MΩ
C) 833 kΩ
D) 750 kΩ
E) 535 kΩ

Chemistry

1A 1	2A 2	3B 3	4B 4	5B 5	6B 6	7B 7	8B 8	8B 9	8B 10	1B 11	2B 12	3A 13	4A 14	5A 15	6A 16	7A 17	8A 18
1 H 1.01																	2 He 4.00
3 Li 6.94	4 Be 9.01											5 B 10.81	6 C 12.01	7 N 14.01	8 O 16.00	9 F 19.00	10 Ne 20.18
11 Na 22.99	12 Mg 24.31											13 Al 26.98	14 Si 28.09	15 P 30.97	16 S 32.07	17 Cl 35.45	18 Ar 39.95
19 K 39.10	20 Ca 40.08	21 Sc 44.96	22 Ti 47.87	23 V 50.94	24 Cr 52.00	25 Mn 54.94	26 Fe 55.85	27 Co 58.93	28 Ni 58.69	29 Cu 63.55	30 Zn 65.38	31 Ga 69.72	32 Ge 72.64	33 As 74.92	34 Se 78.96	35 Br 79.90	36 Kr 83.80
37 Rb 85.47	38 Sr 87.62	39 Y 88.91	40 Zr 91.22	41 Nb 92.91	42 Mo 95.94	43 Tc (98)	44 Ru 101.07	45 Rh 102.91	46 Pd 106.42	47 Ag 107.87	48 Cd 112.41	49 In 114.82	50 Sn 118.71	51 Sb 121.76	52 Te 127.60	53 I 126.90	54 Xe 131.29
55 Cs 132.91	56 Ba 137.33	57 La 138.9	72 Hf 178.49	73 Ta 180.95	74 W 183.84	75 Re 186.21	76 Os 190.23	77 Ir 192.22	78 Pt 195.08	79 Au 196.97	80 Hg 200.59	81 Tl 204.38	82 Pb 207.20	83 Bi 208.98	84 Po (209)	85 At (210)	86 Rn (222)
87 Fr (223)	88 Ra (226)	89 Ac (227)	104 Rf (261)	105 Db (262)	106 Sg (266)	107 Bh (264)	108 Hs (277)	109 Mt (268)	110 Ds (281)	111 Rg (281)	112 Cn (285)	113 Nh (286)	114 Fl (289)	115 Mc (289)	116 Lv (293)	117 Ts (293)	118 Og (294)

58 Ce 140.1	59 Pr 140.9	60 Nd 144.2	61 Pm (145)	62 Sm 150.4	63 Eu 152.0	64 Gd 157.3	65 Tb 158.9	66 Dy 162.5	67 Ho 164.9	68 Er 167.3	69 Tm 168.9	70 Yb 173.0	71 Lu 175.0
90 Th 232.0	91 Pa 231.0	92 U 238.0	93 Np (237)	94 Pu (244)	95 Am (243)	96 Cm (247)	97 Bk (247)	98 Cf (251)	99 Es (252)	100 Fm (257)	101 Md (258)	102 No (259)	103 Lr (262)

Water Data

T_{mp} = 0°C
T_{bp} = 100°C
c_{ice} = 2.09 J/g·K
c_{water} = 4.184 J/g·K
c_{steam} = 2.03 J/g·K
ΔH_{fus} = 334 J/g
ΔH_{vap} = 2260 J/g
K_f = 1.86 °C/m
K_b = 0.512 °C/m

Standard Reduction Potentials

$Cu^{2+} + 2e^- \rightarrow Cu(s)$ $E°_{red} = +0.34$ V
$Al^{3+} + 3e^- \rightarrow Al(s)$ $E°_{red} = -1.66$ V

Constants

$R = 0.08206$ L·atm/mol·K
$R = 8.314$ J/mol·K
$R = 62.36$ L·torr/mol·K
$e = 1.602 \times 10^{-19}$ C
$N_A = 6.022 \times 10^{23}$ mol^{-1}
$k = 1.38 \times 10^{-23}$ J/K
$h = 6.626 \times 10^{-34}$ J·s
$c = 3.00 \times 10^{8}$ m/s
$R_H = 2.178 \times 10^{-18}$ J
$m_e = 9.11 \times 10^{-31}$ kg

Chemistry

1A 1	2A 2	3B 3	4B 4	5B 5	6B 6	7B 7	8B 8	8B 9	8B 10	1B 11	2B 12	3A 13	4A 14	5A 15	6A 16	7A 17	8A 18
1 H 1.01																	2 He 4.00
3 Li 6.94	4 Be 9.01											5 B 10.81	6 C 12.01	7 N 14.01	8 O 16.00	9 F 19.00	10 Ne 20.18
11 Na 22.99	12 Mg 24.31											13 Al 26.98	14 Si 28.09	15 P 30.97	16 S 32.07	17 Cl 35.45	18 Ar 39.95
19 K 39.10	20 Ca 40.08	21 Sc 44.96	22 Ti 47.87	23 V 50.94	24 Cr 52.00	25 Mn 54.94	26 Fe 55.85	27 Co 58.93	28 Ni 58.69	29 Cu 63.55	30 Zn 65.38	31 Ga 69.72	32 Ge 72.64	33 As 74.92	34 Se 78.96	35 Br 79.90	36 Kr 83.80
37 Rb 85.47	38 Sr 87.62	39 Y 88.91	40 Zr 91.22	41 Nb 92.91	42 Mo 95.94	43 Tc (98)	44 Ru 101.07	45 Rh 102.91	46 Pd 106.42	47 Ag 107.87	48 Cd 112.41	49 In 114.82	50 Sn 118.71	51 Sb 121.76	52 Te 127.60	53 I 126.90	54 Xe 131.29
55 Cs 132.91	56 Ba 137.33	57 La 138.9	72 Hf 178.49	73 Ta 180.95	74 W 183.84	75 Re 186.21	76 Os 190.23	77 Ir 192.22	78 Pt 195.08	79 Au 196.97	80 Hg 200.59	81 Tl 204.38	82 Pb 207.20	83 Bi 208.98	84 Po (209)	85 At (210)	86 Rn (222)
87 Fr (223)	88 Ra (226)	89 Ac (227)	104 Rf (261)	105 Db (262)	106 Sg (266)	107 Bh (264)	108 Hs (277)	109 Mt (268)	110 Ds (281)	111 Rg (281)	112 Cn (285)	113 Nh (286)	114 Fl (289)	115 Mc (289)	116 Lv (293)	117 Ts (293)	118 Og (294)

58 Ce 140.1	59 Pr 140.9	60 Nd 144.2	61 Pm (145)	62 Sm 150.4	63 Eu 152.0	64 Gd 157.3	65 Tb 158.9	66 Dy 162.5	67 Ho 164.9	68 Er 167.3	69 Tm 168.9	70 Yb 173.0	71 Lu 175.0
90 Th 232.0	91 Pa 231.0	92 U 238.0	93 Np (237)	94 Pu (244)	95 Am (243)	96 Cm (247)	97 Bk (247)	98 Cf (251)	99 Es (252)	100 Fm (257)	101 Md (258)	102 No (259)	103 Lr (262)

Water Data

T_{mp} = 0°C
T_{bp} = 100°C
c_{ice} = 2.09 J/g·K
c_{water} = 4.184 J/g·K
c_{steam} = 2.03 J/g·K
ΔH_{fus} = 334 J/g
ΔH_{vap} = 2260 J/g
K_f = 1.86 °C/m
K_b = 0.512 °C/m

Standard Reduction Potentials

$Cu^{2+} + 2e^- \rightarrow Cu(s)$ $E°_{red} = +0.34$ V
$Al^{3+} + 3e^- \rightarrow Al(s)$ $E°_{red} = -1.66$ V

Constants

$R = 0.08206$ L·atm/mol·K
$R = 8.314$ J/mol·K
$R = 62.36$ L·torr/mol·K
$e = 1.602 \times 10^{-19}$ C
$N_A = 6.022 \times 10^{23}$ mol^{-1}
$k = 1.38 \times 10^{-23}$ J/K
$h = 6.626 \times 10^{-34}$ J·s
$c = 3.00 \times 10^{8}$ m/s
$R_H = 2.178 \times 10^{-18}$ J
$m_e = 9.11 \times 10^{-31}$ kg

Physics

Useful Constants

quantity	symbol	value
Free-fall acceleration	g	$9.80\ m/s^2$
Permittivity of Free Space	ε_0	$8.854 \times 10^{-12}\ C^2/Nm^2$
Permeability of Free Space	μ_0	$4\pi \times 10^{-7}\ Tm/A$
Coulomb constant	k	$8.99 \times 10^{9}\ Nm^2/C^2$
Speed of light in a vacuum	c	$3.00 \times 10^{8}\ m/s$
Fundamental charge	e	$1.602 \times 10^{-19}\ C$
Planck's constant	h	$6.626 \times 10^{-34}\ Js$
Electron mass	m_e	$9.11 \times 10^{-31}\ kg$
Proton mass	m_p	$1.67265 \times 10^{-27}\ kg$ $1.007276 amu$
Neutron mass	m_n	$1.67495 \times 10^{-27}\ kg$ $1.008665 amu$
Atomic Mass Unit	amu	$1.66 \times 10^{-27}\ kg$ $931.5\ MeV/c^2$
Gravitational constant	G	$6.67 \times 10^{-11}\ Nm^2/kg^2$
Stefan-Boltzmann constant	σ	$5.67 \times 10^{-8}\ W/m^2K^4$
Universal gas constant	R	$8.314\ J/mol \cdot K$ $0.082057\ L \cdot atm/mol \cdot K$
Boltzmann's constant	k_B	$1.38 \times 10^{-23}\ J/K$
Speed of Sound (at 20ºC)	v	343 m/s
Avogadro's number	N_A	$6.022 \times 10^{23}\ atoms/mol$
Electron Volts	eV	$1.602 \times 10^{-19}\ J/eV$
Distance Conversion	miles → meters	1.00 mile = 1609 meters
Rydberg Constant	R_∞	$1.097 \times 10^{7}\ m^{-1}$
Standard Atmospheric Pressure	1 atm	$1.013 \times 10^{5}\ Pa$
Density of Pure Water	ρ_{water}	$1000.0\ kg/m^3$